

4.4 STARTING ENGINES

- NOTE**
- At ambient temperatures below +4 °C, refer to “Cold weather starting information” this section.
 - Either engine may be started first.
 - If, for any reason, a starting attempt is discontinued the entire starting sequence must be repeated from the beginning.
 - When a starting attempt is aborted but TOT limits were not exceeded, wait 15 seconds after N_1 has returned to zero before attempting restart to permit excess fuel to drain from combustion chamber.
 - If a start is not successful after three attempts, maintenance action is required.
 - Before starting engines, the operational hydraulic system shall always be pre-selected to system 2 (collective lever lock engaged).

Fire guard (if available)	– Posted
Rotor area	– Clear
Anti-collision light	– On

4.4.1 Starting First Engine

NOTE Operate only the controls of the corresponding engine.

CAUTION IMMEDIATELY ABORT START FOR ANY OF THE FOLLOWING:

- MAIN ROTOR IS NOT ROTATING AT 25% N_1 . (A SECOND AND THIRD START ATTEMPT MAY BE MADE. IF POWER TURBINE STILL FAILS TO ROTATE, DISCONTINUE ATTEMPT).
- STAGNATED START (FUEL STARVATION).
- TOT RISES ABNORMALLY AND RAPIDLY APPROACHES START LIMIT (927 °C).
- TOT EXCEEDS 10-SECOND LIMIT WITHIN TEMPERATURE RANGE (811-927 °C).
- NO POSITIVE ENGINE OR TRANSMISSION OIL PRESSURE INDICATIONS UPON REACHING IDLE SPEED (N_1 59-65%).
- IDLE SPEED IS NOT REACHED WITHIN 60 SECONDS UNLESS N_1 AND N_2 ARE ACCELERATING AND TOT IS WITHIN LIMITS.
- N_2 INCREASES BEYOND N_{R0} . A FREEWHEELING UNIT SLIPPAGE IS SUSPECTED AND MAINTENANCE ACTION IS REQUIRED.
- AUTO-ACCELERATION (BENDIX FUEL CONTROL SYSTEM INSTALLED). SEE “COLD WEATHER STARTING INFORMATION” THIS SECTION.

EFFECTIVITY Hydraulic Systems **without** Positive Cross-over Indication

- EITHER **HYD 1** OR **HYD 2** WARNING LIGHT REMAINS ON UPON REACHING GROUND IDLE SPEED (60% N_{R0}).

EFFECTIVITY Hydraulic Systems *with* Positive Cross-over Indication

- **HYD 2** WARNING LIGHT REMAINS ON UPON REACHING GROUND IDLE SPEED (60% N_{R0}).

EFFECTIVITY ALL

ABORT START PROCEDURE

- POWER LEVER – OFF
- FUEL PUMPS – OFF
- IGNITION SWITCH – OFF
- STARTER SWITCH – ON, AS NECESSARY TO LOWER TOT
- STARTER SWITCH – OFF

- TOT – Less than 150 °C, otherwise ventilate engine
- Fuel pumps (4) – On
- Ignition switch – On
- Starter switch – On; start clock
- N_1 – Check increasing

CAUTION POWER LEVER MUST NOT BE MOVED FROM OFF POSITION UNTIL REQUIRED N_1 HAS BEEN REACHED. STARTING AT N_1 SPEEDS LESS THAN 12% INCREASES THE POSSIBILITY OF EXCEEDING ENGINE TOT LIMITS.

As N_1 accelerates through 12-15%:

If CECO fuel control installed:

- Power lever – Slowly advance to a position slightly before IDLE (trim start)

If BENDIX fuel control installed:

- Power lever – Move direct to a position slightly before IDLE

NOTE Even small displacements of the power lever from IDLE to OFF position may cause engine to flameout.

NOTE Do not wait for N_1 to peak out. Move power lever towards IDLE immediately upon reaching the desired N_1 speed. Delay in moving power lever may diminish battery capacity early in the start cycle.

- TOT and N_1 – Monitor closely, observe TOT peak
- Power lever – Lock in IDLE
- N_2/N_{R0} – Check increasing